Ambient Care Companion

A hardware + AI/ML + full-stack capstone project for elderly monitoring, reminders, and companionship — no physical robot, delivered entirely through a smart home hub and mobile apps.

Inspiration: This plan adapts ideas from the GUARDIAN ecosystem (Ciuffreda et al., "Design and Development of a Technological Platform Based on a Sensorized Social Robot for Supporting Older Adults and Caregivers," International Journal of Social Robotics, 2025) — a European research project that paired a social robot with caregiver/senior mobile apps around three needs: **monitoring, reminding, and social companionship.** We keep that three-pillar structure but replace the commercial robot (Misty II) with a custom ESP32-based smart hub, since our brief is a hardware + software build that fits a 3-person team with AI/ML, full-stack, and embedded skills.

Team Split (1 line each)

Member	Primary Owns	Also Touches
AI/ML	Companion chat, wellbeing NLU, anomaly/routine detection	Reminder scheduling logic
Full-Stack	Senior app, caregiver app/dashboard, backend API, DB	Push notifications, auth
Hardware	ESP32 hub firmware, sensors, audio I/O, MQTT publish	Device provisioning flow

1. Problem Statement & Vision

Older adults living alone face three recurring risks: missed medication/meals, loneliness, and caregivers not knowing something is wrong until it's serious. Our system should give caregivers **visibility** (monitoring), give seniors **structure** (reminders), and give seniors **someone to talk to** (companionship) — all through a low-cost hub placed in one room of the house plus two mobile app experiences.

We are deliberately **not** building a robot, and we are **not** centering the project on fall detection. Instead the differentiators are: (a) a genuinely useful AI companion layer instead of a scripted reminder bot, and (b) a hardware hub that senses the environment and routine, not just the person's body.

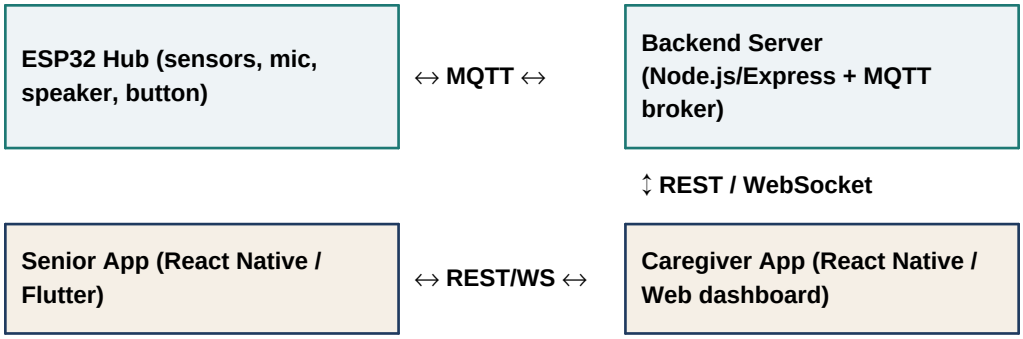
2. System Overview

Three components mirror the paper's ecosystem, mapped onto what a 3-person team can realistically ship in one final-year cycle:

- **Smart Home Hub (Hardware)** — ESP32-based device with sensors, mic, and speaker. Sits in the senior's living room. Detects presence, speaks reminders, listens for check-in replies, and has one big physical button for manual check-ins/help.
- **Senior Companion App** — large-text, low-friction mobile app mirroring what the hub says, for seniors who prefer tapping over talking. Doubles as a fallback if the hub is out of range.
- **Caregiver App** — dashboard for one or more caregivers: colour-coded status timeline, reminder/medication setup, live environment snapshot, and alerts.

All three talk to a central backend, which also hosts the AI/ML services (chat, speech-to-text/text-to-speech orchestration, anomaly detection). This backend is the one piece every member will touch, so its API contract should be agreed and frozen early.

Architecture at a glance



The AI/ML services (chat model, STT/TTS, anomaly detection) run as modules inside or alongside the backend and are called by it — they don't talk to the hub or apps directly. This keeps the hardware and app contracts simple: they only ever speak to one backend.

3. Features — MVP (build this first, in this order)

3.1 Reminders (backbone feature — build first)

- Caregiver sets medication / meal / appointment reminders via the caregiver app (time, frequency, label).
- At the scheduled time, backend pushes the reminder to the hub (spoken aloud) and to the senior app (shown on screen) simultaneously.
- Senior responds by voice ("done"/"not yet") or by tapping a button on the hub/app.
- Response is logged and shown to the caregiver as a coloured status (done / missed / no response) on a timeline — same idea as GUARDIAN's dashboard dot legend.

3.2 Monitoring

- Hub reports periodic presence/activity signal (PIR motion + sound level) so the caregiver app can show "active since 8am" / "no activity detected in living room for 4 hrs".
- Environment snapshot: temperature, humidity, and (optionally) air quality shown on the caregiver dashboard.
- Self-reports: senior periodically answers short prompts ("How are you feeling?", "Did you eat?", "How did you sleep?") via voice or app taps; caregiver sees these on a timeline exactly like the reminder responses.
- Routine-deviation alerts: if the senior's daily pattern (wake time, first activity, meal self-reports) departs noticeably from their own historical baseline, flag it for the caregiver. This is the AI/ML team's anomaly-detection deliverable — framed around routine, not framed as fall detection.

3.3 Companionship

- Hub can hold a short spoken conversation (greet the senior, answer simple questions, chat about the weather/day) using an LLM API for the dialogue, wrapped so replies stay short, friendly, and non-repetitive — the paper's user requirements table (Table 2) is a good checklist for exactly this: short messages, variety, always end on a positive note.

- Compliments/encouragement triggered by self-report data (e.g., a light "good job taking a walk today" after a positive activity self-report).
- A big physical "talk to me / help" button on the hub as the lowest-friction way to start an interaction — important for seniors who won't touch a touchscreen.

3.4 Core platform (not visible as a "feature" but required for all the above)

- Auth + caregiver-senior linking (one senior can have multiple caregivers).
- Push notifications (Firebase Cloud Messaging) for caregiver alerts.
- Device provisioning: hub connects to home Wi-Fi (BLE or captive-portal setup) and pairs to a senior's account with a QR code / pairing code.

4. Stretch Goals (add only once the MVP above works end-to-end)

Feature	Owner	Why it's a good add-on
Wake-word / hands-free listening on the hub (instead of push-to-talk)	Hardware + AI/ML	Makes voice interaction feel natural; doable with a lightweight on-device keyword spotter.
Personalised daily menu / activity suggestions via RAG on senior's stated preferences	AI/ML	Reuses your existing RAG/LangChain experience; matches GUARDIAN's "personalised suggestion" requirement.
Sleep-quality estimate from night-time sound level + light sensor pattern	Hardware + AI/ML	New signal beyond self-report; feeds the monitoring dashboard.
Emergency escalation (auto-SMS/call a caregiver if a check-in gets no response twice)	Full-Stack	High perceived value for caregivers, relatively low build cost (Twilio-style API).
Multi-caregiver chat / notes thread on a shared senior profile	Full-Stack	Matches GUARDIAN's caregiver-collaboration requirement; good UX depth.
Bilingual voice support (e.g. Hindi + English) for reminders and chat	AI/ML	Real-world relevance for deployment in India; strong differentiator for a jury.
Gas/air-quality safety alert (MQ-series sensor) separate from wellbeing monitoring	Hardware	Cheap to add once the sensor bus exists; genuine safety value, not just elder-care specific.

Pick 2–3 of these, not all — the jury rewards a working, polished MVP far more than a long unfinished feature list.

5. Technology Stack

Layer	Suggested tech	Owner
Hub firmware	ESP32 (Arduino/ESP-IDF), PubSubClient (MQTT), I2S mic (INMP441) + I2S amp (MAX98357A), PIR sensor, DHT22, small OLED/e-ink, push button, RGB LED	Hardware
Device → cloud transport	MQTT (Mosquitto/HiveMQ) for sensor data & commands; HTTPS for firmware/audio upload if needed	Hardware + Full-Stack
Backend	Node.js + Express (matches your MERN experience), MQTT client, WebSocket (Socket.IO) for live dashboard updates	Full-Stack
Database	MongoDB (reminders, self-reports, sensor logs, users)	Full-Stack
Senior app / Caregiver app	React Native (single codebase, two role-based flows) or Flutter	Full-Stack
Voice pipeline	STT + TTS via an API (e.g. existing Groq/Gemini access) orchestrated by the backend; hub only streams/plays raw audio	AI/ML + Hardware
Companion chat	LLM API call (Gemini/Groq) with a short system prompt enforcing brevity, warmth, and variety, grounded with the senior's stored profile	AI/ML
Anomaly / routine detection	Classical ML to start (rolling baseline + isolation forest / z-score on daily activity & self-report features); upgrade to a small sequence model only if time allows	AI/ML
Push notifications	Firebase Cloud Messaging	Full-Stack
Hosting	Any free-tier cloud VM or Render/Railway for backend + broker; MongoDB Atlas free tier	Full-Stack

6. Data Flow — worked example (medication reminder)

- Caregiver app: caregiver adds "Paracetamol, 10:20am daily" → saved to backend DB.
- Backend scheduler fires at 10:20 → checks the hub last saw the senior nearby (recent PIR/mic activity) → publishes an MQTT message to the hub's reminder topic and pushes the same payload to the senior app.
- Hub speaks the reminder via TTS; senior app shows the same text with Yes/No buttons.
- Senior replies (voice → hub records → backend runs STT → intent "taken"/"not taken", or a direct app tap).
- Backend logs the response, updates the caregiver dashboard timeline with a green/red/amber dot for that slot, and sends a push notification only if the reminder was missed.

This is exactly the flow described for GUARDIAN's medication reminders (Section 4.4 of the paper) — we're keeping the same logic, just swapping "query RDS at the pre-set time" for "backend cron/scheduler", and swapping the robot's speaking skill for the hub's TTS playback.

7. Suggested Timeline (adjust to your semester calendar)

Phase	Duration	Deliverable
1. Requirements & architecture	1–2 weeks	Finalised feature list (this doc, trimmed), API contract, sensor list, wireframes
2. Parallel skeleton build	3–4 weeks	Hub publishes dummy sensor data over MQTT; backend has auth + reminder CRUD; both apps have empty-state screens hitting real APIs
3. Feature build-out	4–5 weeks	Reminders end-to-end; monitoring dashboard live; companion chat working on the hub
4. Integration + hardening	2 weeks	All three components talking to the same backend reliably; error/edge-case handling (hub offline, missed reminders, etc.)
5. Usability testing + polish	2 weeks	Test with 5–8 real users (see §8); fix top issues; record demo
6. Report + stretch goals	remaining time	Add 1–2 stretch features if ahead of schedule; write the report

8. Evaluation Plan

GUARDIAN's authors validated their system with real older adults, informal caregivers, and formal caregivers using the IBM Computer Usability Satisfaction Questionnaire plus open-ended interview questions (their Tables 7–9). We can't run a multi-country study, but a small-scale version of the same idea will strengthen the final report a lot:

- Recruit 5–8 people across two roles: a few older adults (grandparents, family friends) and a few informal caregivers (parents, relatives who look after them).
- Run a 20–30 minute session per person: free exploration, then 2–3 set tasks (set a reminder, respond to a reminder, check the dashboard).
- Use a short usability questionnaire (5–8 items on a 1–5 scale is enough for a student project, e.g. "it was easy to use," "I felt comfortable," "I would use this").
- Ask 2–3 open questions afterward: what did you like, what confused you, what's missing.
- Report both the quantitative scores and a couple of direct quotes — this is exactly the mixed-methods format examiners expect and mirrors the paper's own results structure.

9. Risks & Mitigations

Risk	Mitigation
Hub Wi-Fi/MQTT drops mid-demo	Add a visible connection-status LED/icon (GUARDIAN did this with green/yellow/red icons after facing the same issue); auto-reconnect logic on the ESP32
Voice pipeline (STT/TTS) is slow or unreliable	Always keep the app UI as a fallback path for every voice interaction; don't make voice the only way to respond
Scope creep from the stretch-goal list	Freeze the MVP feature list after Phase 1 and treat everything else as optional, ranked by owner and effort
One member's component blocks the others	Agree the API contract (endpoints + MQTT topics/payloads) in Phase 1 so all three can build against mocks in parallel
Real elderly test users are hard to recruit	Start recruiting in Phase 1, not Phase 5; family/neighbours are a fine substitute for a student project

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Reference: Ciuffreda, I. et al. (2023/2025). "Design and Development of a Technological Platform Based on a Sensorized Social Robot for Supporting Older Adults and Caregivers: GUARDIAN Ecosystem." *International Journal of Social Robotics*, 17, 803–822. https://doi.org/10.1007/s12369-023-01038-5